

Data Science

An Introduction

Setia Pramana

Data, data and data everywhere.....

Big Data is affecting people everywhere.



Big Data is changing business



Big data industry estimated to be worth **\$100 BILLION+**

IDC believes that within five years..... companies will finally be able to run a real-time enterprise that simultaneously transacts, analyzes and acts on big data.



SAP is helping customers get real value from Big Data

MKI performs genome analysis with SAP HANA

"...with this (SAP HANA) we've found a way to shorten the genomic analysis time from several days down to **only 20 minutes.**"

YUKIHISA KATO, CTO AND DIRECTOR OF MKI

eBay uses predictive analytics to gain new insights

"With the **speed of HANA** great people become exceptional at what they do because of the **speed** that they can **interact** with the data. That is truly **awesome.**"

DANIEL SCHWARZBACH, VP & CFO EBAY NORTH AMERICA AT EBAY INC.

Bigpoint solves big data challenges with SAP HANA

Our expectation – and it actually seems to be coming true – is that the use of this technology and the methods behind it helps us realize **sales growth spurts of 10 – 30%.**

MICHAEL GUTSMANN, CFO BIGPOINT

Find out what opportunities Big Data holds for you. Follow us on Twitter @asiarunbetter Visit www.sap.com/bigdata Contact SAP

Gartner: The Nexus of Forces Boosts Information Governance and MDM, 2012 | IDC: IDC Predictions 2012: Competing for 2020 | IDC: Ingraining Insights into the DNA of People and Process, 2013 | International Telecommunications Union, 2013 | Economist, 2013 | SAP, 2013



THE WORLD OF DATA

NUMBER
OF EMAILS
SENT
EVERY SECOND

2.9
MILLION

DATA
CONSUMED BY
HOUSEHOLDS
EACH DAY

375
MEGABYTES

VIDEO
UPLOADED TO
YOUTUBE EVERY
MINUTE

20
HOURS

DATA PER
DAY
PROCESSED
BY GOOGLE

24
PETABYTES

TWEETS
PER
DAY

50
MILLION

TOTAL MINUTES
SPENT ON
FACEBOOK
EACH MONTH

700
BILLION

DATA SENT
AND RECEIVED
BY MOBILE
INTERNET USERS

1.3
EXABYTES

PRODUCTS
ORDERED ON
AMAZON PER
SECOND

72.9
ITEMS



IN THE 21ST CENTURY, we live a large part of our lives online. Almost everything we do is reduced to bits and sent through cables around the world at light speed. But just how much data are we generating? This is a look at just some of the massive amounts of information that human beings create every single day.

© 2014 BBC. Data sources: Apple, Google, Amazon, Facebook, Twitter, etc.

How Large is Big?

Byte of data : one grain of rice

Kilobyte : cup of rice

Megabyte : 8 bags of rice

Gigabyte : 3 container lorries

Terabyte : 2 container ships

Petabyte : covers Manhattan

Exabyte : covers the UK (3 times)

Zettabyte : fills the pacific ocean

Yottabyte : Earth size rice ball

Type of Data

- Relational Data (Tables/Transaction/Legacy Data)
- Text Data (Web)
- Semi-structured Data (XML)
- Graph Data
- Social Network, Semantic Web (RDF), ...
- Streaming Data
- You can afford to scan the data once

What to do?

- Aggregation and Statistics
 - Data warehousing and OLAP
- Indexing, Searching, and Querying
 - Keyword based search
 - Pattern matching (XML/RDF)
- Knowledge discovery
 - Data Mining
 - Statistical Modeling



Analytics Approaches

- Descriptive: What happened or what is happening now?
- Diagnostic: Why did it happen or Why is it happening now?
- Predictive: What will happen next? What will happen under various conditions?
- Prescriptive: What are the options to create the most optimal/high value result/outcome?

Data Science

“Applying advanced statistical tools to existing data to solve problems, generate new insights, improve products/services”

“Everything that has something to do with data: Collecting, analyzing, modeling..... yet the most important part is its applications --- all sorts of application”

What is Data Science?

- Theories and techniques from many fields and disciplines are used to investigate and analyze a large amount of data to help decision makers in many industries such as science, engineering, economics, politics, finance, and education
 - Computer Science
 - Pattern recognition, visualization, data warehousing, High performance computing, Databases, AI
 - Mathematics
 - Mathematical Modeling
 - Statistics
 - Statistical and Stochastic modeling, Probability.

Data Science: An Action Plan for Expanding the Technical Areas of the Field of Statistics

William S. Cleveland

Statistics Research, Bell Labs

DOI:10.1002/sam.11239

Published online in Wiley Online Library (wileyonlinelibrary.com).

Abstract: An action plan to expand the technical areas of statistics focuses on the data analyst. The plan sets out six technical areas of work for a university department and advocates a specific allocation of resources devoted to research in each area and to courses in each area. The value of technical work is judged by the extent to which it benefits the data analyst, either directly or indirectly. The plan is also applicable to government research labs and corporate research organizations. © 2014 Wiley Periodicals, Inc. *Statistical Analysis and Data Mining* 7: 414–417, 2014

1. SUMMARY OF THE PLAN

This article describes a plan to broaden the major areas of technical work of the field of statistics. Because the plan is ambitious and implies substantial changes, the plan will be called ‘data science’.

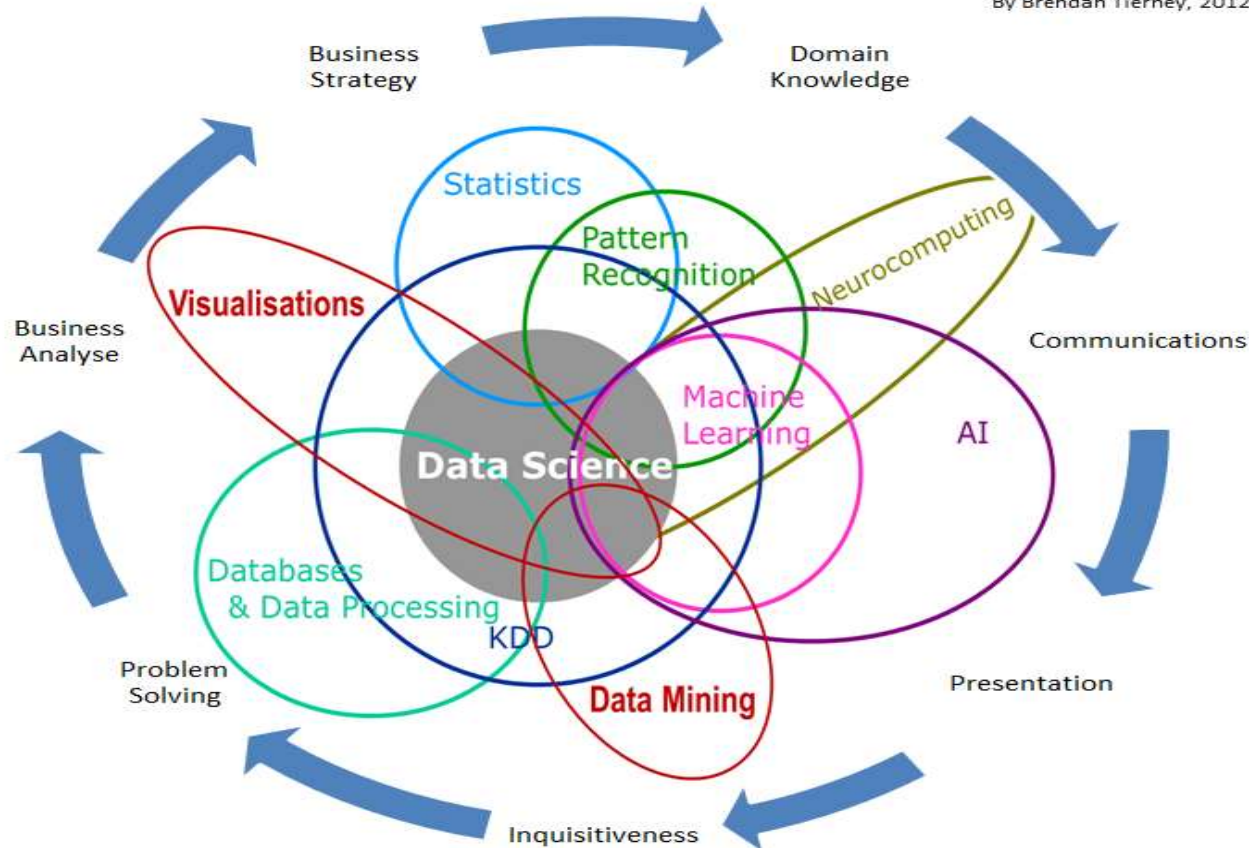
The focus of the plan is broadening the data analyst. A

- *Models and Methods for Data (20%)*: statistical models; methods of model building; and methods of estimation and distribution based on probabilistic

hardware systems;

Data Science Is Multidisciplinary

By Brendan Tierney, 2012



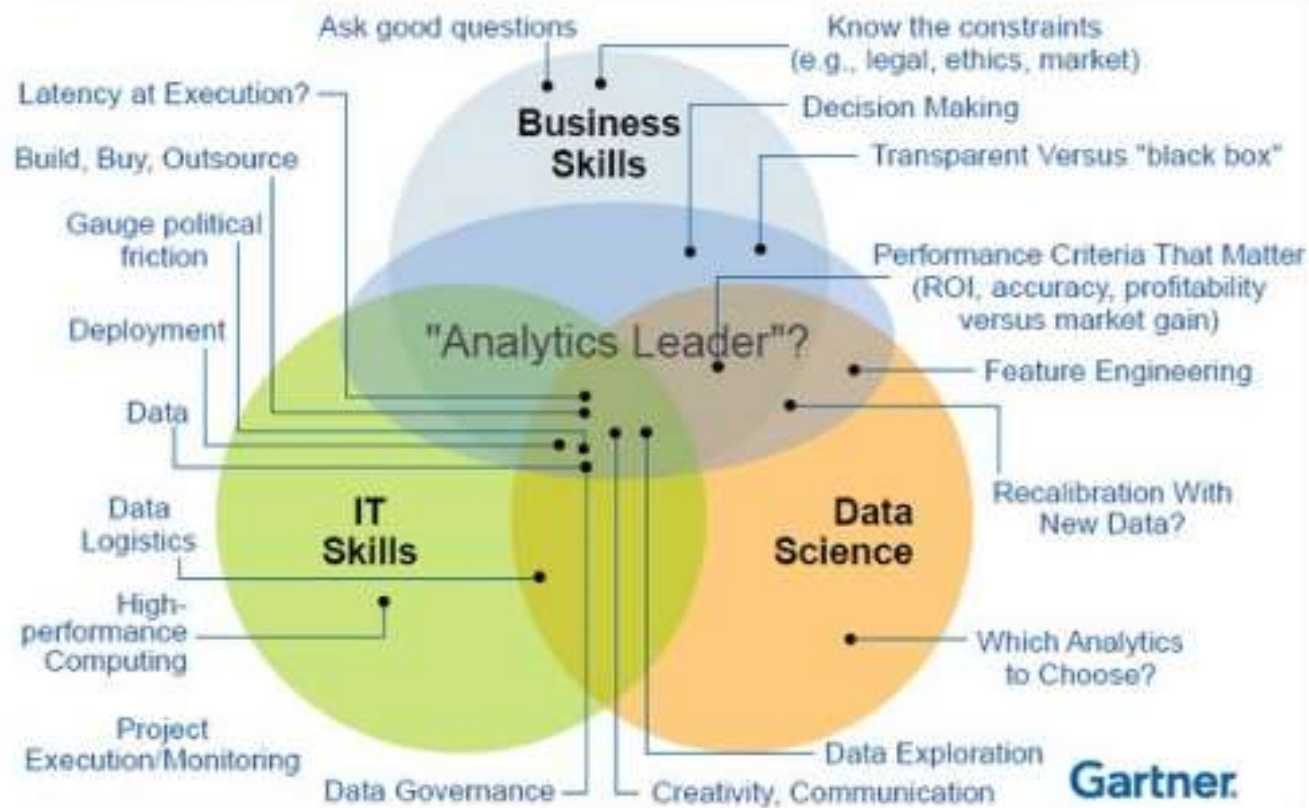
Data Science

- A Mashed Up Discipline
- A multi-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data

Math and Theory	• Statistics, Linear Algebra, Optimization, Time Series, etc.
Applied Algorithms	• Machine Learning, Data Structures, Parallel Algorithms, etc.
Engineering and Technologies	• Storage and computing platforms, statistical tools ,etc.
Domain Expertise	• Text, Finance, Images, Econometrics etc.
Art	• Visualization, Infographics
Best practices and hacks	• Handle missed values in data, transform and represent data, etc.

Data Science

Driving the Success of Data Science Solutions: Skills, Roles and Responsibilities ...



ANATOMY OF A DATA SCIENTIST

SALARY

Average salary of data scientists is **\$120,000/year**

BENEFITS

- Harvard Business Review called data science the **"Sexiest Job of the 21st Century"**
- One of the fastest growing careers in the United States
- 94%** of data science graduates have found jobs since 2011

RESPONSIBILITIES

- Conduct research
- Extract, clean, and analyze data from varied sources
- Solve problems
- Build automation tools
- Communicate findings to management



EDUCATION

- 88%** of all data scientists have at least a Master's degree
- 46%** of data scientists have a PhD

SKILLS

- Programming languages (R, Python, SQL, Hive, etc.)
- Statistics
- Multivariable calculus and linear algebra
- Machine learning
- Software engineering
- Wrangle, visualize, and communicate data to management

CAREER POSSIBILITIES

- The majority of data scientists work in the **technology industry**.
- Other options include marketing, consulting, healthcare and pharmaceuticals, finance, government, gaming, and many more.

RESOURCES:

<https://insidebigdata.com/2017/08/05/benefits-data-scientist-career/>
https://www.glassdoor.com/Salaries/us-data-scientist-salary_SRCH_IL_Q2_IN1_KO3,17.htm
<https://blog.udacity.com/2014/11/data-science-job-skills.html>



THE COMPUTER MERCHANT, LTD.
THE IT TRAINING COMPANY

MODERN DATA SCIENTIST

Data Scientist, the sexiest job of 21st century requires a mixture of multidisciplinary skills ranging from an intersection of mathematics, statistics, computer science, communication and business. Finding a data scientist is hard. Finding people who understand who a data scientist is, is equally hard. So here is a little cheat sheet on who the modern data scientist really is.

MATH & STATISTICS

- Machine learning
- Statistical modeling
- Experiment design
- Bayesian inference
- Supervised learning: decision trees, random forests, logistic regression
- Unsupervised learning: clustering, dimensionality reduction
- Optimization: gradient descent and variants

PROGRAMMING & DATABASE

- Computer science fundamentals
- Scripting language e.g. Python
- Statistical computing package e.g. R
- Databases SQL and NoSQL
- Relational algebra
- Parallel databases and parallel query processing
- MapReduce concepts
- Hadoop and Hive/Pig
- Custom reducers
- Experience with xaaS like AWS

DOMAIN KNOWLEDGE & SOFT SKILLS

- Passionate about the business
- Curious about data
- Influence without authority
- Hacker mindset
- Problem solver
- Strategic, proactive, creative, innovative and collaborative

COMMUNICATION & VISUALIZATION

- Able to engage with senior management
- Story telling skills
- Translate data-driven insights into decisions and actions
- Visual art design
- R packages like ggplot or lattice
- Knowledge of any of visualization tools e.g. Flare, D3.js, Tableau



MarketingDistillery.com is a group of practitioners in the area of e-commerce marketing. Our fields of expertise include: marketing strategy and optimization; customer tracking and on-site analytics; predictive analytics and econometrics; data warehousing and big data systems; marketing channel insights in Paid Search, SEO, Social, CRM and brand.

Marketing
DISTILLERY

THE DATA SCIENCE **HIERARCHY OF NEEDS**

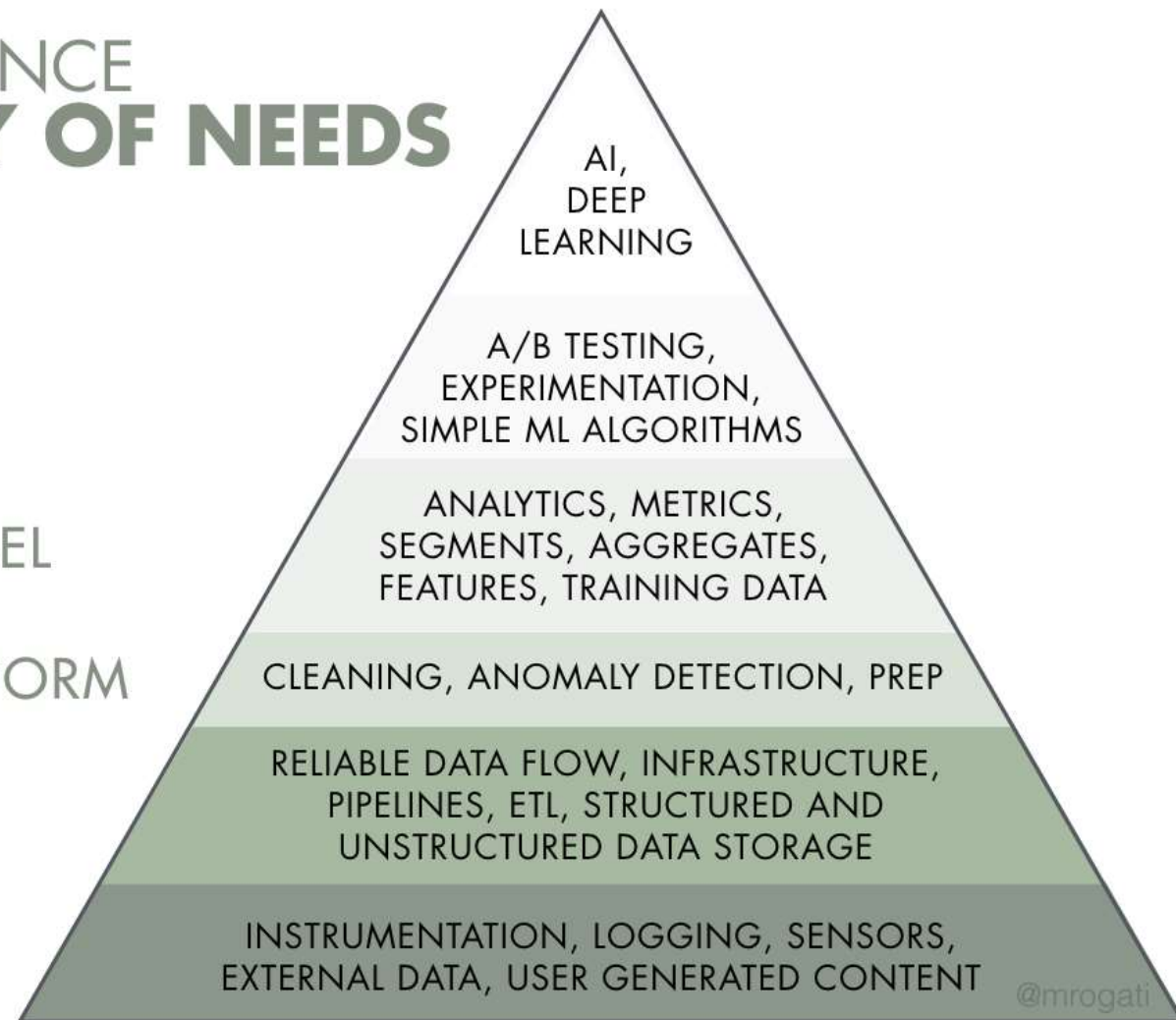
LEARN/OPTIMIZE

AGGREGATE/LABEL

EXPLORE/TRANSFORM

MOVE/STORE

COLLECT



THE DATA SCIENCE HIERARCHY OF NEEDS

Start Up

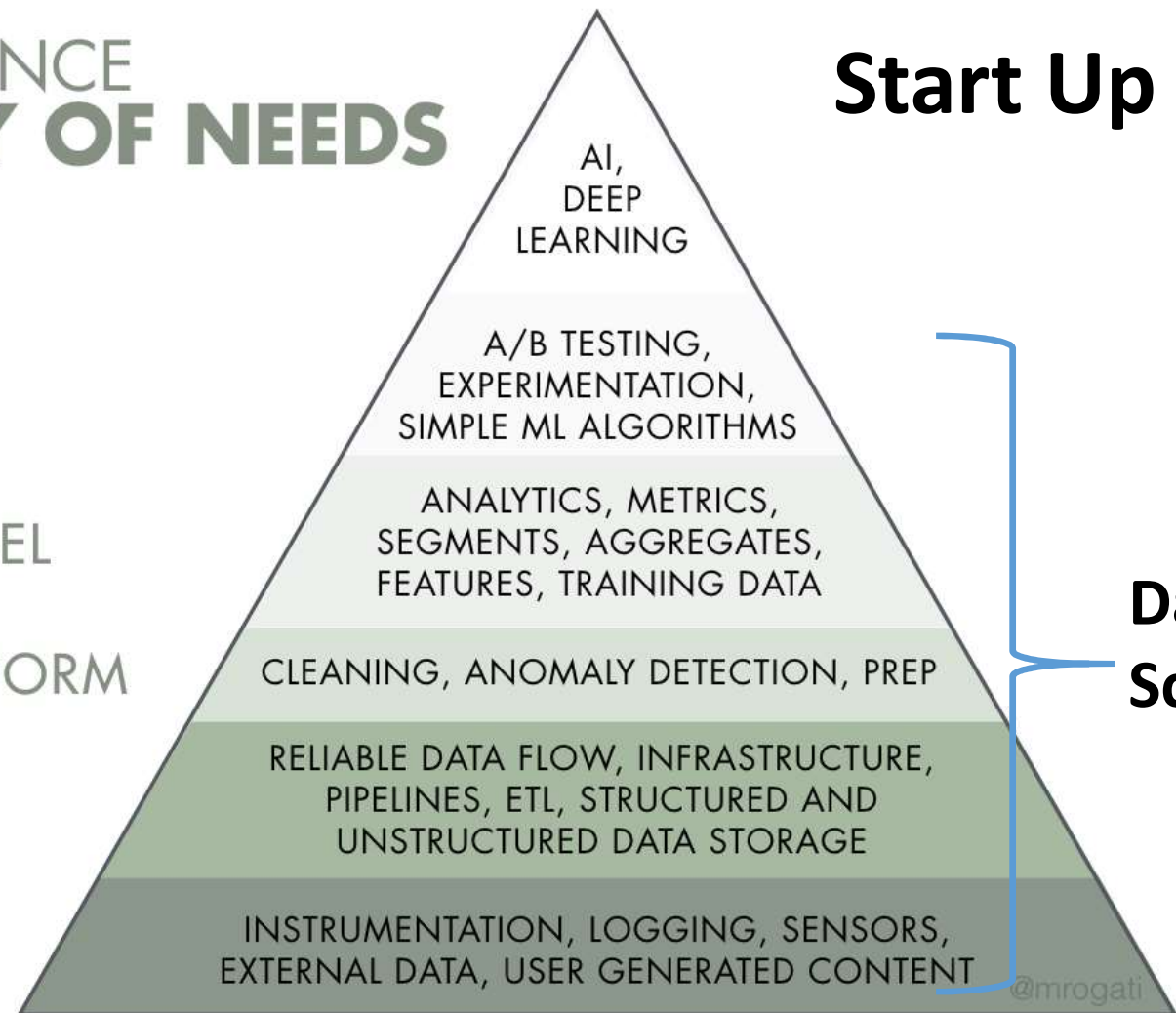
LEARN/OPTIMIZE

AGGREGATE/LABEL

EXPLORE/TRANSFORM

MOVE/STORE

COLLECT



THE DATA SCIENCE HIERARCHY OF NEEDS

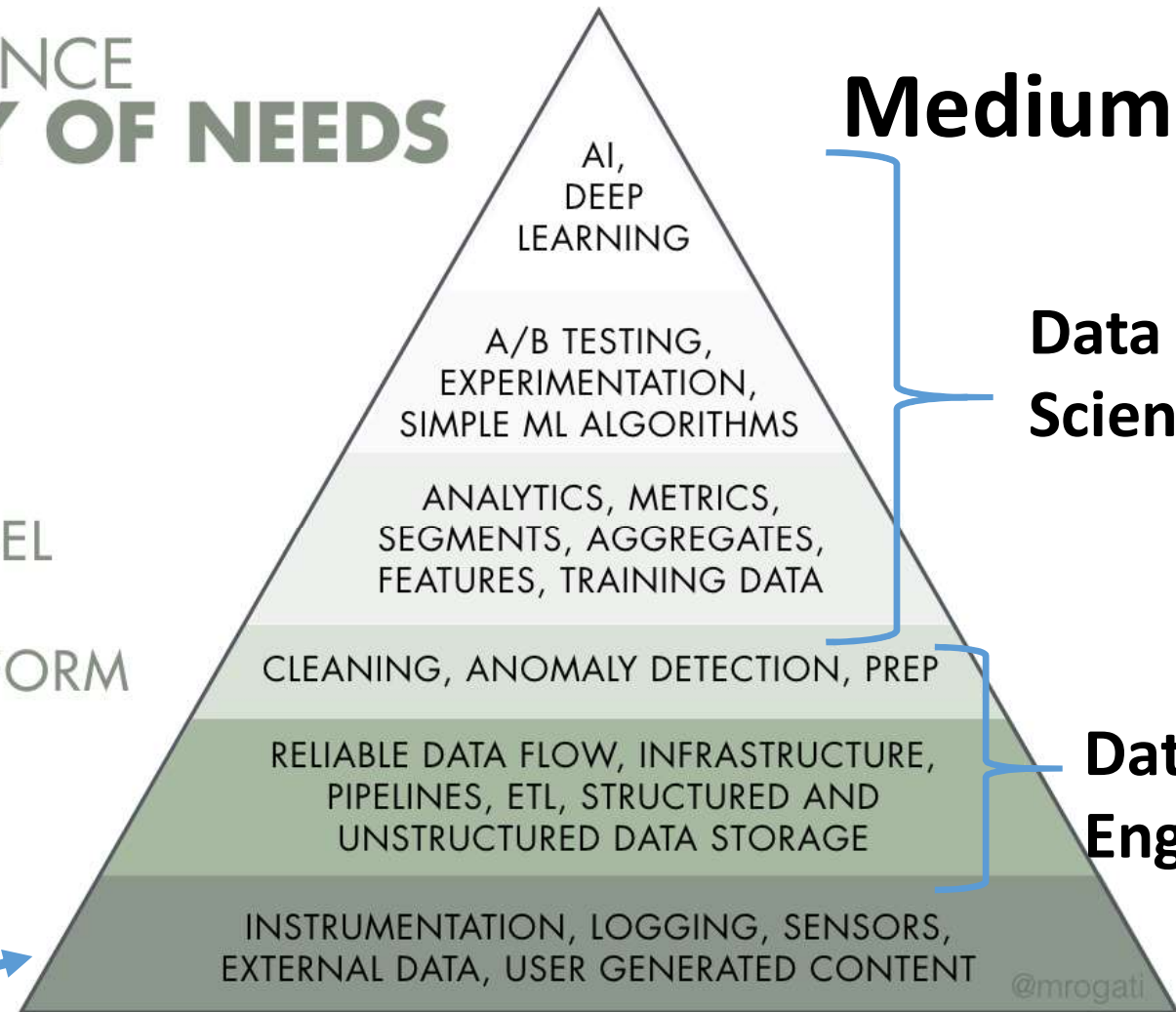
LEARN/OPTIMIZE

AGGREGATE/LABEL

EXPLORE/TRANSFORM

MOVE/STORE

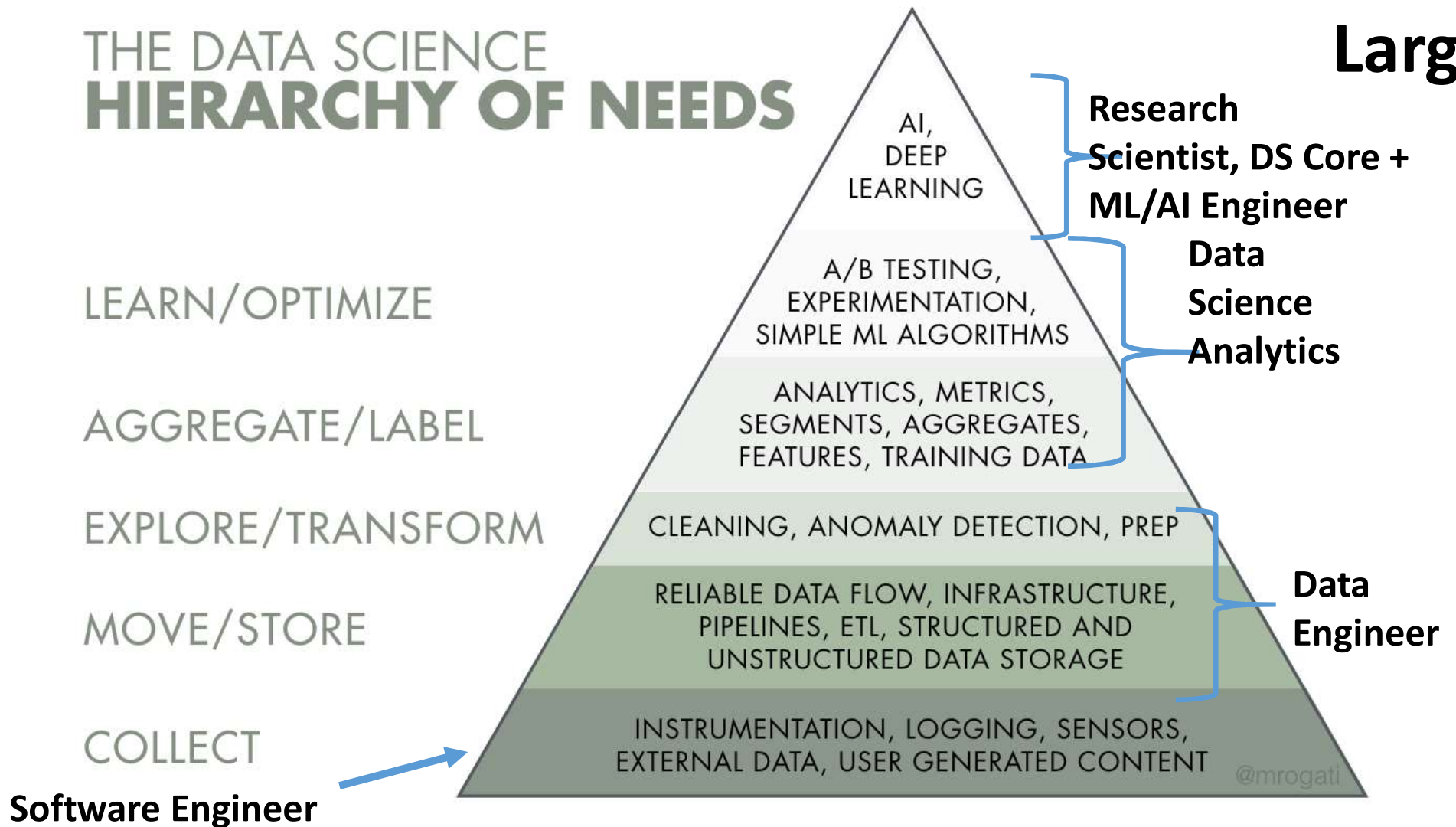
COLLECT



Software Engineer

THE DATA SCIENCE HIERARCHY OF NEEDS

Large



DATA Engineer



DataCamp
Learn Data Science By Doing

Develops, constructs, tests, and maintains architectures. Such as databases and large-scale processing systems.

DATA Scientist

Cleans, massages and organizes (big) data. Performs descriptive statistics and analysis to develop insights, build models and solve a business need.



Languages, Tools & Software



Source: datacamp

Data Scientist

also known as Data Managers, statisticians.



A data scientist will be able to take data science projects from end to end. They can help store large amounts of data, create predictive modelling processes and present the findings.

Skills: Mathematics, Programming, Communication



Will use programmes such as:
SQL, Python, R

Data Engineers

also known as database administrators and data architects.



They are versatile generalists who use computer science to help process large datasets. They typically focus on coding, cleaning up data sets, and implementing requests that come from data scientists.

Skills: Programming, Mathematics, Big data



Will use programmes such as:
Hadoop, NoSQL, and Python

Data Analysts

also known as business Analysts.



They typically help people from across the company understand specific queries with charts.

Skills: Statistics, Communication, Business knowledge



Will use programmes such as:
Excel, Tableau, SQL

Data Scientist: The Sexiest Job of the 21st Century

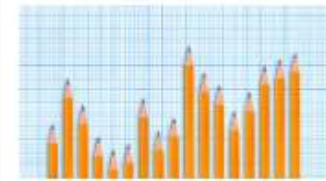
by [Thomas H. Davenport](#) and [D.J. Patil](#)

FROM THE OCTOBER 2012 ISSUE

 Summary  Save  Share  ¹⁶ Comment  **H** Text Size  Print **\$8.95** Buy Copies

When Jonathan Goldman arrived for work in June 2006 at LinkedIn, the business networking site, the place still felt like a start-up. The company had just under 8 million accounts, and the number was growing quickly as existing members invited their friends and colleagues to join. But users weren't seeking out connections with the people who were already on the site at the rate executives had expected. Something was apparently missing in the social

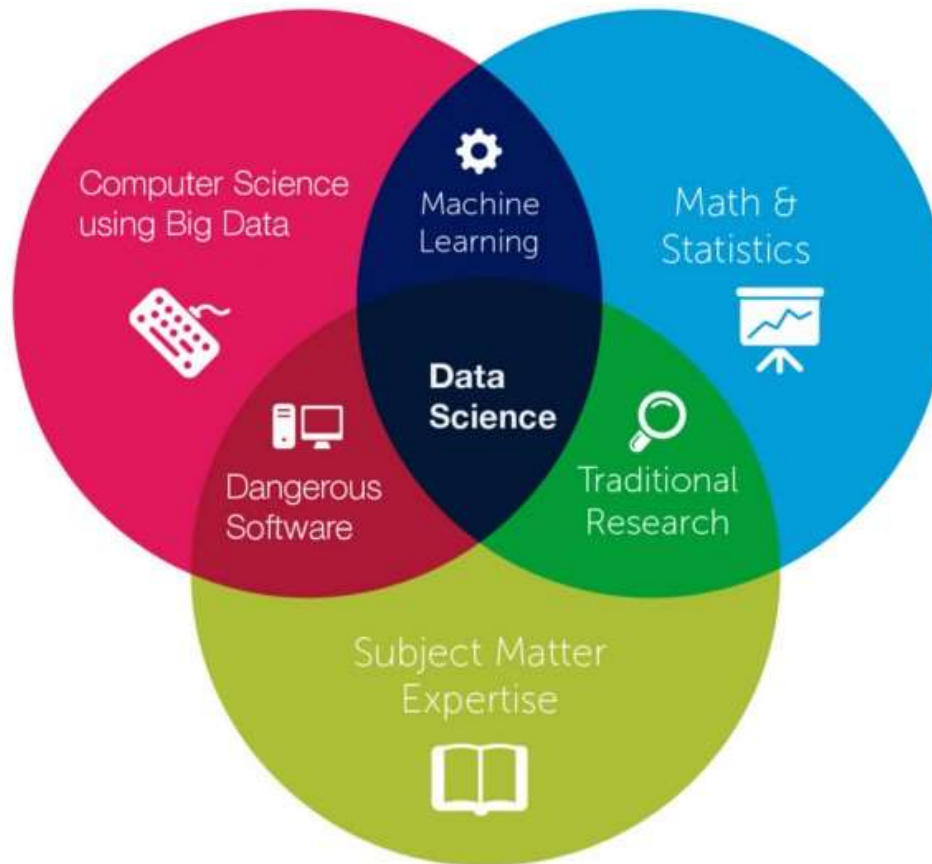
WHAT TO READ NEXT



What Data Scientists Really Do,
According to 35 Data Scientists

VIEW MORE FROM THE
October 2012 Issue





Methods

Sentiment analysis

Time series analysis

Data mining

Multilevel modeling

Missing data imputations

Classification and clustering

Survival analysis

Pattern recognition

Principal component and factor analysis

AB testing

Machine learning

Forecasting

Propensity score matching

Logistic, multinomial and multiple linear regression techniques

Network analysis

Tools

Languages

Python

R

SQL

Javascript

NodeJS

Libraries

SciPy

Pandas

Scikit-learn

GPText

OpenNLP

Mahout

+many others

Data Engineering

Profiling

ETL

Job notices

APIs

Optimized data
pipelines

Optimized data
storage/access

Visualization

D3.js

Gephi

R

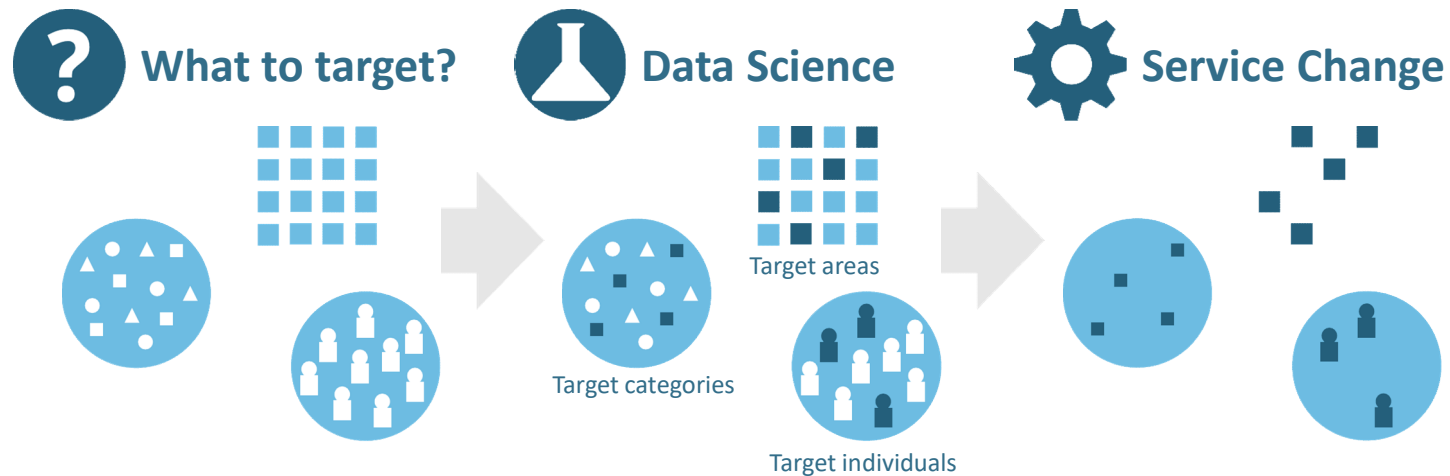
Leaflet

PowerBI

ggplot2

shiny

Targeting: Find the needle in the haystack



Service Issue:
Difficult to identify
targets in a population

Data Science Process:
Use existing data and
predictive modeling to
identify targets

Service Change:
Engage with target
subset of population

Result: Department resources are spent where most needed

Prioritizing



Service Issue:

Backlog is tackled via first in, first out (FIFO)

Data Science Process:

Create a model to categorize and group past and current cases

Service Change:

Prioritize cases based on categories in order of risk, need or opportunity

Result: Department addresses high priority cases first

Predictions



Service Issue:

Hard to predict future condition which leads to reactive services

Data Science Process:

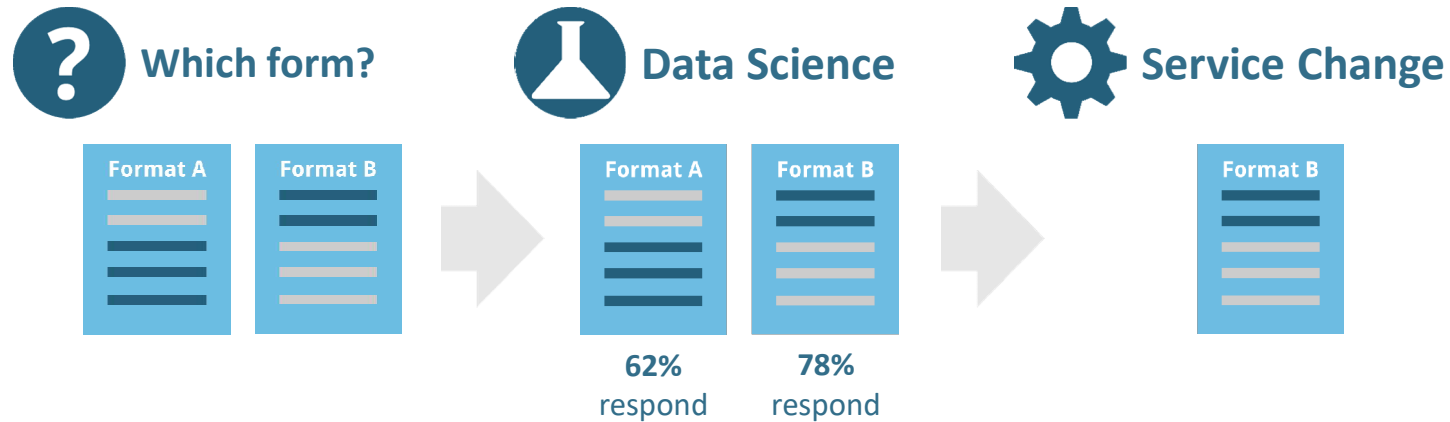
Use historical and current data to create estimate ranges for potential outcomes

Service Change:

Use estimates to change and tailor intervention points

Result: Department provides pro-active early interventions

A/B test



Service Issue:

Costly outreach methods are not tested before implementation

Data Science Process:

Statistical testing on outreach methods to identify which, when, and to whom to send

Service Change:

Use statistically validated outreach method

Result: Department increases response rates

Optimization



Service Issue:

Difficult to identify where to place or distribute resources to be most effective

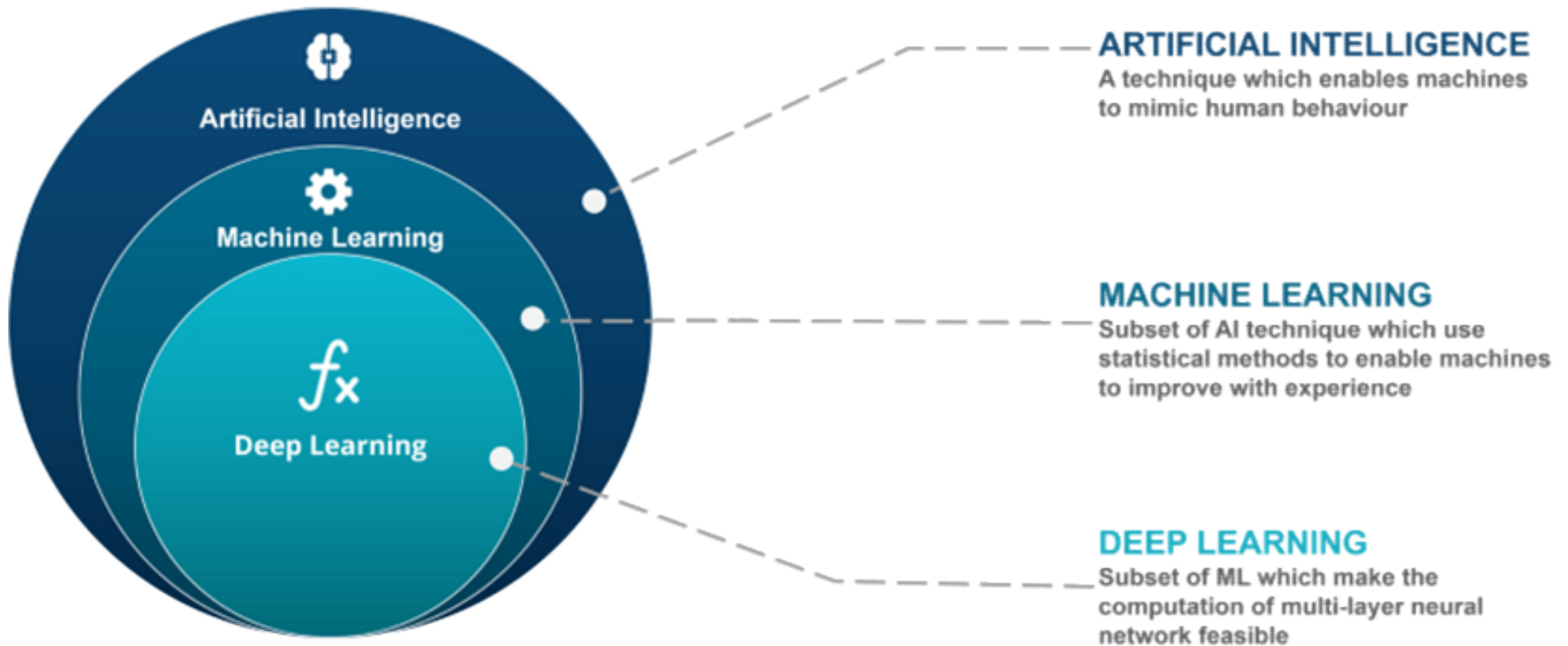
Data Science Process:

Use geospatial and/or other data to identify optimal distribution of resources

Service Change:

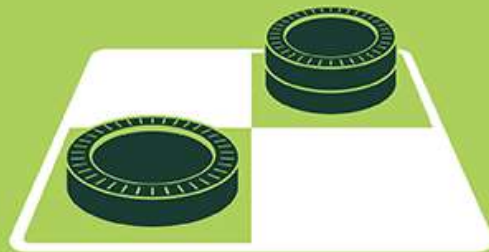
Re-allocates resources to optimal distribution

Result: Department decreases response times; increases volume



ARTIFICIAL INTELLIGENCE

Early artificial intelligence stirs excitement.



MACHINE LEARNING

Machine learning begins to flourish.



DEEP LEARNING

Deep learning breakthroughs drive AI boom.



1950's

1960's

1970's

1980's

1990's

2000's

2010's

Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

<https://www.datasciencecentral.com> 32

Data Mining, AI and Machine Learning

- **Data Mining:** extract existing information to highlight patterns, and serves as foundation for AI and machine learning.
- **Artificial Intelligence:** creating machines that perform functions that require intelligence when performed by people.
- **Machine Learning:** Offers data necessary for a machine to learn & adapt. The machine must automatically learn the parameters of models from the data. It uses self-learning algorithms to improve its performance at a task with experience over time

AI in Sci-Fi Movies

- Terminator



<http://starwars.com/>

- Iron Man Marvel



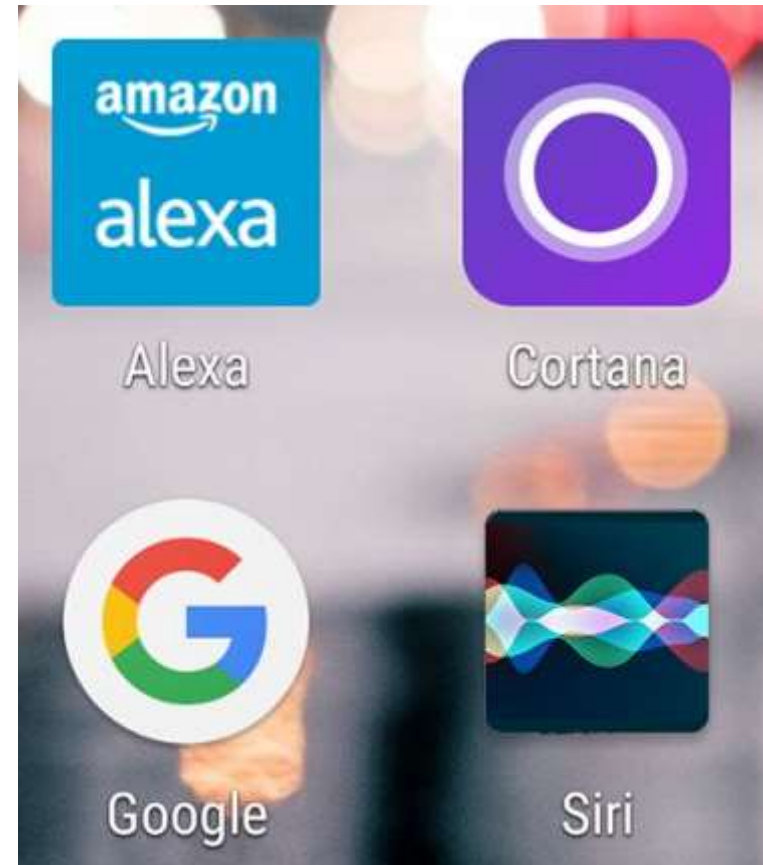
Just A Rather Very Intelligent System

AI in Life

Vacuum Cleaning Robot



AI assistants



AI in Life

- Kiva warehouse robot



What is Artificial Intelligence ?

- The art of creating machines that perform functions that require intelligence when performed by people (Kurzweil, 1990)
- The study of how to make computers do things at which, at the moment, people are better (Rich and Knight, 1991)
- AI: acting humanly

Machine Learning

Learning from
Experience



Learning from
Data

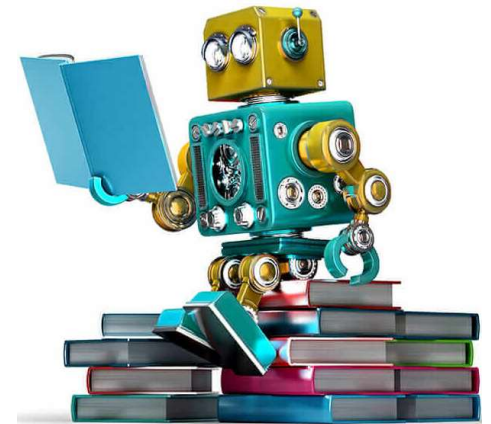


Follow
Instructions



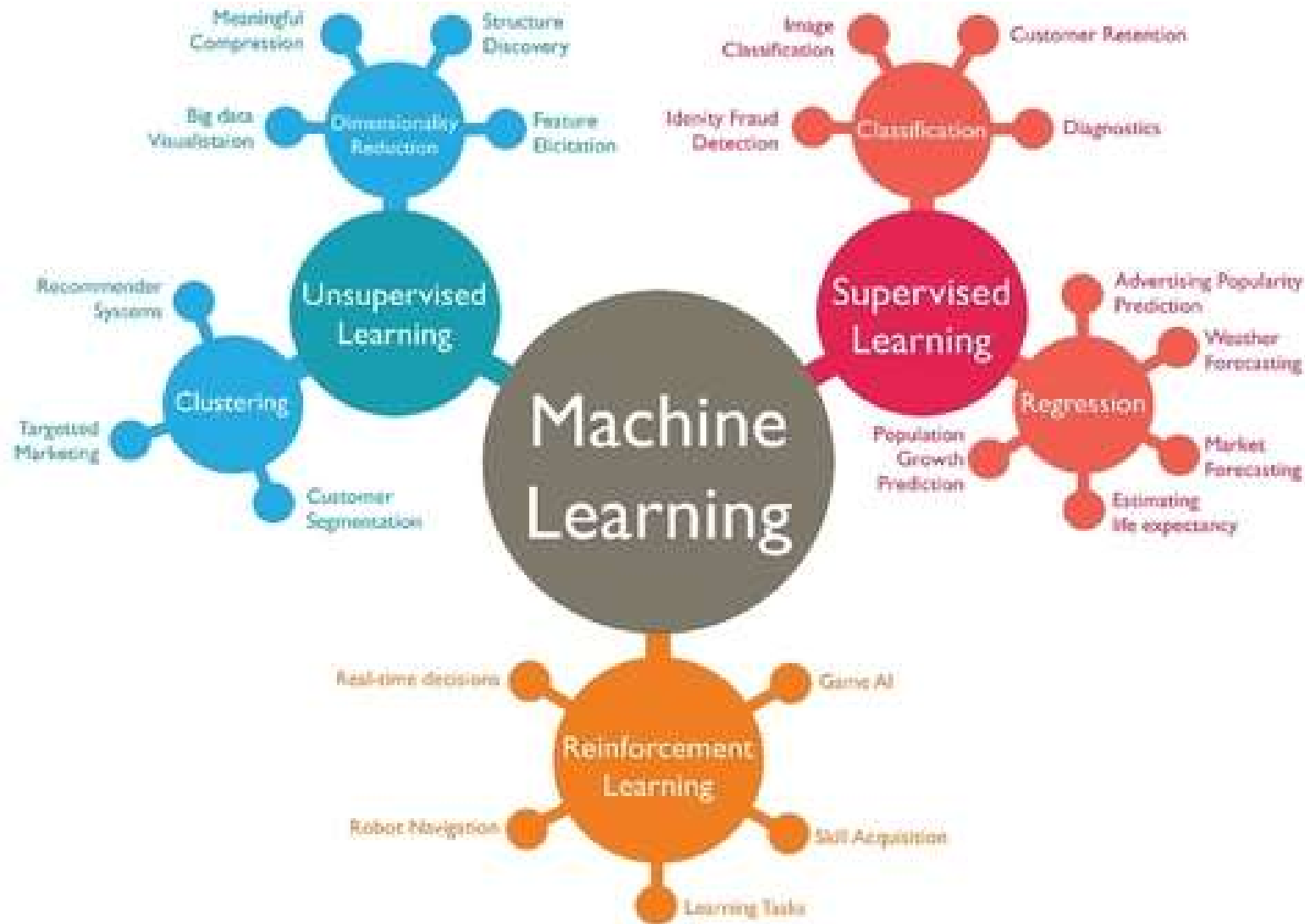
Machine Learning

- Machine learning is aimed to optimize a certain task using example data or past experience
- The extraction of knowledge from data
- Machine learning is preferred approach to
 - Business Intelligence
 - Speech recognition, Natural language processing
 - Computer vision
 - Robot control
 - Computational biology
 - Crime predictions
 - Etc..



Types of Learning

- **Supervised (inductive) learning**
 - Training data includes desired outputs
- **Unsupervised learning**
 - Training data does not include desired outputs
- **Reinforcement learning**
 - Rewards from sequence of actions



Methods

- **Supervised learning**

- Decision tree induction
- Rule induction
- Naïve Bayes
- Neural networks
- Support vector machines
- Model ensembles
- Etc.

- **Unsupervised learning**

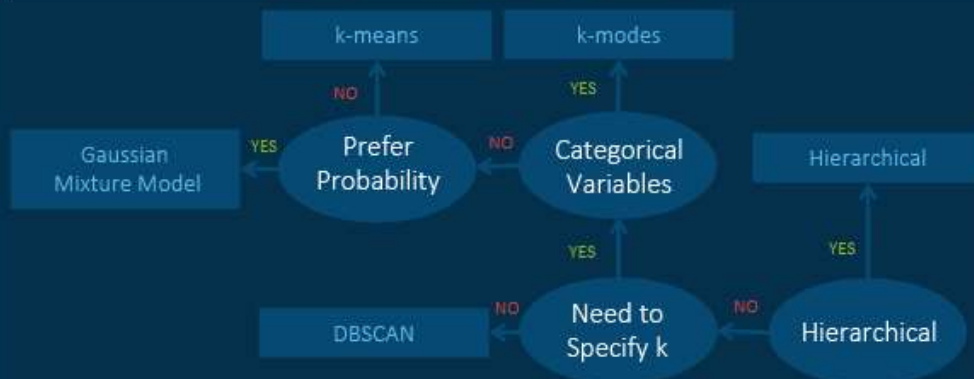
- Clustering
- Dimensionality reduction

- **Reinforcement learning**

- Decision making (robot, chess machine)

Machine Learning Algorithms Cheat Sheet

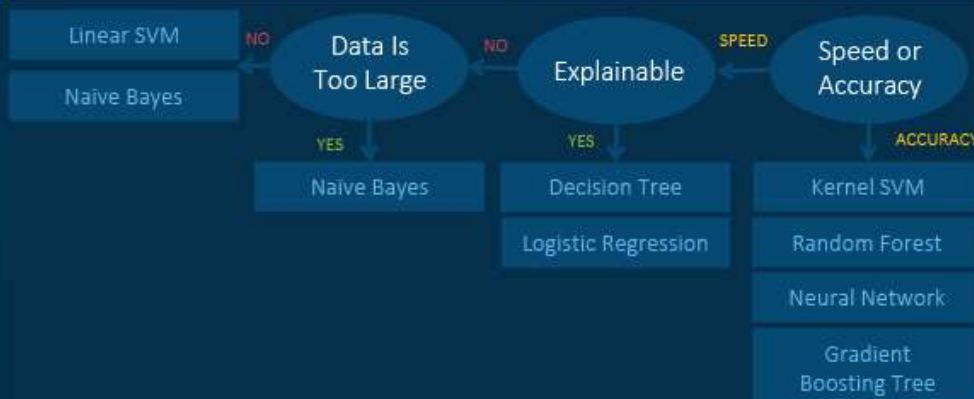
Unsupervised Learning: Clustering



Unsupervised Learning: Dimension Reduction



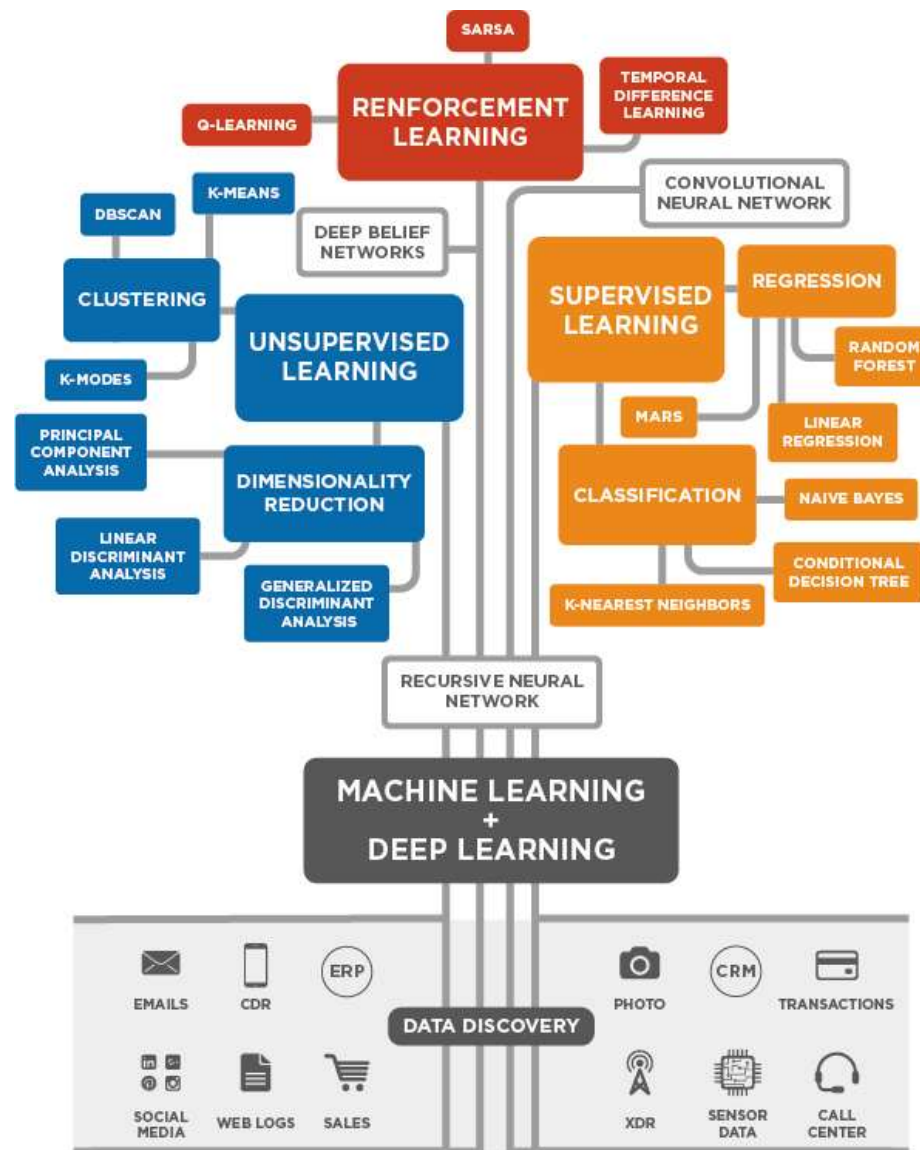
Supervised Learning: Classification



Supervised Learning: Regression



Numerous (New) Algorithms

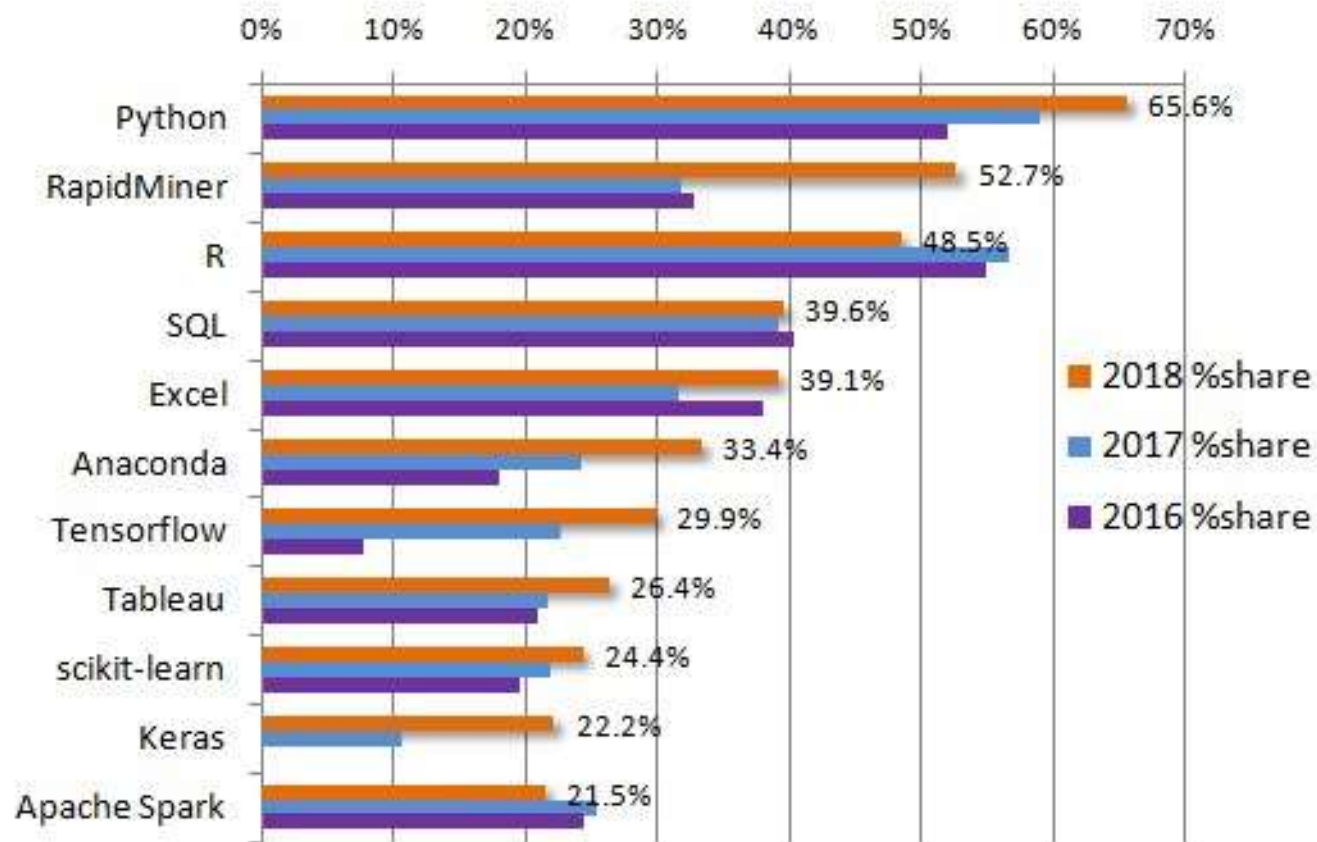


Numerous (New) Algorithms

Classification algorithms considered in the benchmarking study.

	BM selection	Classification algorithm	Acronym					
Individual classifier	n.a.	Bayesian Network	B-Net	Heterogeneous ensembles	n.a.	Simple average ensemble Weighted average ensemble Stacking	AvgS AvgW Stack	
		CART	CART		Static direct	Complementary measure Ensemble pruning via reinforcement learning GASEN	CompM EPVRL GASEN	
		Extreme learning machine	ELM			Hill-climbing ensemble selection HCES with bootstrap sampling	HCES HCES-B	
		Kernalized ELM	ELM-K			Matching pursuit optimization ensemble	MPOE	
		k-nearest neighbor	kNN			Top-T ensemble	Top-T	
		J4.8	J4.8		Static indirect	Clustering using compound error k-Means clustering	CuCE k-Mean	
		Linear discriminant analysis ²	LDA			Kappa pruning	KaPru	
		Linear support vector machine	SVM-L			Margin distance minimization	MDM	
		Logistic regression ²	LR			Uncertainty weighted accuracy	UWA	
		Multilayer perceptron artificial neural network	ANN		Dynamic	Probabilistic model for classifier competence k-nearest oracle	PMCC kNORA	
		Naive Bayes	NB					
		Quadratic discriminant analysis ²	QDA					
		Radial basis function neural network	RbfNN					
		Regularized logistic regression	LR-R		Classification models from heterogeneous ensembles			
		SVM with radial basis kernel function	SVM- Rbf		17			
		Voted perceptron	VP					
Classification models from individual classifiers			16					
Homogenous ensembles	n.a.	Alternating decision tree	ADT					
		Bagged decision trees	Bag					
		Bagged MLP	BagNN					
		Boosted decision trees	Boost					
		Logistic model tree	LMT					
		Random forest	RF					
		Rotation forest	RotFor					
		Stochastic gradient boosting	SGB					

KDnuggets Analytics, Data Science, Machine Learning Software Poll, 2016-2018



Thank you!

setia.pramana@stis.ac.id

